AGV- DRONE COLLABORATIVE 3D PRINTING SYSTEM FOR RAPID TEMPORARY SHELTER CONSTRUCTION

Project report submitted
in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Mechatronics and Automation

by

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SCHOOL OF MECHANICAL ENGINEERING
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CHENNAI

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DECLARATION

We, Mohammad Safwan (22BMH1105), Velan K (22BMH1091) and Vikash Madhava (22BMH1083), B Tech Mechatronics and Automation Engineering, VIT Chennai, hereby declare that the work being presented in the dissertation entitled **"AGV- DRONE COLLABORATIVE 3D PRINTING SYSTEM FOR RAPID TEMPORARY SHELTER CONSTRUCTION"** is an authentic record of the work that has been carried out at VIT, Chennai under the guidance of Dr.Arockia Selvakumar, School of Mechanical Engineering, VIT Chennai.

We hereby declare that the entire work embodied in this dissertation has been carried out by us and no part of it has been submitted for any degree or diploma of any institution previously.

Place: Chennai

Date: 08/ 04/ 26

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CERTIFICATE

This is to certify that the project report entitled **"AGV- DRONE COLLABORATIVE 3D PRINTING SYSTEM FOR RAPID TEMPORARY SHELTER CONSTRUCTION"** submitted by Mohammad Safwan (22BMH1105), Velan K (22BMH1091) and Vikash Madhava (22BMH1083), for the award of the degree of *Bachelor of Technology* is a record of bonafide work carried out by him/her/them under my supervision, as per the VIT code of academic and research ethics. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The report fulfils the requirements and regulations of VIT and in my opinion meet the necessary standards for submission.

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ABSTRACT

Our project focuses on designing and implementing a reliable actuation system for material dispensing in a collaborative aerial–ground 3D-printing platform. The main objective was to enable controlled actuation of the solenoid actuator to dispense the material in layers. The quadcopter drone was equipped with a pressurized canister for lightweight expandable foam extrusion is utilized to deposit the material in a controlled manner. A key challenge identified was the incompatibility between the Pixhawk's PWM based AUX outputs and the solenoid's requirement for a clean digital ON/OFF signal. Direct interfacing led to issues such as partial actuation, voltage leakage and inconsistent retraction of the actuator.

The proposed system consists of two main subsystems: an AGV designed for wireless ground movement and potential material support and a quadcopter drone developed for controlled material dispensing. The AGV was implemented using an ESP32-based control system with a web interface for real-time operation. The drone was assembled using an S500 frame and configured using a Pixhawk 2.4.8 flight controller with QGroundControl. A lightweight material dispensing mechanism using a canister and solenoid valve was developed and tested. Due to practical limitations, continuous material transfer from AGV to drone was not implemented. The dispensing mechanism was successfully validated through ground-level testing.

Although stable flight with payload was not achieved, the project demonstrates the feasibility of key subsystems and identifies critical challenges such as payload limitations and system stability. The developed prototype provides a foundation for future work toward a fully integrated AGV–drone collaborative construction system.

Keywords: Pixhawk solenoid interface, MOSFET low-side switching, AGV-drone collaboration, Aerial 3D printing, Drone material dispensing, Temporary shelter

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LIST OF NOMENCLATURE

Symbol	Description	Unit
Vb	Battery voltage supplied to UAV	V
Ib	Battery current drawn by system	A
Vout	Buck converter output voltage	V
Is	Solenoid coil current	A
Rs	Solenoid coil resistance	Ω
ton	Solenoid ON time (valve open)	s
toff	Solenoid OFF time (valve release)	s
Ps	Solenoid power consumption	W
Es	Energy per actuation cycle	J
VGS	MOSFET gate-source voltage	V
ID	MOSFET drain current	A
RDS(on)	MOSFET on-resistance	Ω

LIST OF ACRONYMS AND KEYWORDS

UAV – Unmanned Aerial Vehicle

AGV – Automated Guided Vehicle

PWM – Pulse Width Modulation

ESC – Electronic Speed Controller

IMU – Inertial Measurement Unit

GPS – Global Positioning System

RC – Radio Control

MOSFET – Metal Oxide Semiconductor Field Effect Transistor

DC – Direct Current

BLDC – Brushless Direct Current (Motor)

FPV – First Person View

QGC – QGroundControl

PID – Proportional Integral Derivative (Controller)

Li-Po – Lithium Polymer (Battery)

CPU – Central Processing Unit

IoT – Internet of Things

CAD – Computer Aided Design

RDS(on) – Drain-Source On Resistance

KEYWORDS

Pixhawk Interface, Solenoid Actuation, Drone Construction, AGV Collaboration, PWM Conditioning

CHAPTER 1

INTRODUCTION

1.1 Background

The integration of autonomous robotics with additive manufacturing represents a transformative approach to construction, particularly for rapid shelter deployment in disaster zones and remote regions. Traditional methods face many challenges, skilled labour shortages and infrastructure limitations, especially in India where temporary housing needs are acute. Aerial additive manufacturing using drones offers mobility and rapid deployment while ground-based Automated Guided Vehicles (AGVs) provide structural stability.

This project bridges these domains through a hybrid drone-AGV system for layered material deposition. Recent literature focuses on high-level path planning and material science, but practical actuation interfaces receive limited attention. Commercial drone spraying systems use proprietary controllers, while open-source UAV research rarely addresses solenoid valve integration with flight controllers. This creates a critical gap between conceptual aerial 3D printing demonstrations and field-deployable hardware. The hybrid drone-AGV systems requires precise material dispensing synchronized with coordinated navigation.

1.2 Problem Statement

Existing construction solutions for temporary shelters are not optimized for rapid deployment in unstructured and emergency environments. Large-scale automated systems, such as gantry-based 3D printers are typically stationary and require significant setup, limiting their applicability in disaster-affected or remote locations. On the other hand, drone-based approaches offer flexibility and accessibility but are constrained by limited payload capacity, short flight duration and stability challenges during operation.

The integration of multiple robotic systems for collaborative construction introduces additional complexities, including reliable communication, synchronization and efficient material handling. These factors make it difficult to implement fully functional automated systems in real-world emergency scenarios where speed, adaptability, and reliability are critical. There exists a gap between advanced conceptual systems and their practical implementation at a prototype level. Therefore, this project focuses on developing a simplified system to investigate the feasibility of combining an Autonomous Ground Vehicle (AGV) and a drone for assisted material deployment. Our aim is to evaluate subsystem performance, identify key limitations, and provide a foundation for future development of integrated collaborative construction systems.

1.3 Objective of the Project

The proposed solution involves the development of an integrated aerial-ground system capable of performing controlled material deposition for rapid construction applications. The system consists of a UAV equipped with a pressurized material canister and an electronically actuated solenoid valve, enabling precise regulation of material flow during flight. The actuation mechanism is interfaced with the flight controller to allow real-time control via remote input, ensuring responsive and accurate dispensing aligned with the UAV's position and trajectory.

To address the challenges associated with signal compatibility and reliable actuation, a dedicated driver circuit was implemented to convert control signals into stable switching actions for the solenoid. This ensures consistent performance under varying operational conditions and prevents unintended actuation or material leakage. The dispensing system is further supported by a power management unit to maintain stable voltage levels for both control and actuation components. The autonomous ground vehicle (AGV) is deployed as a mobile support unit that follows the drone, carrying additional material payloads and enabling extended operation without frequent manual intervention. The AGV is to be programmed to maintain a safe and consistent distance from the UAV while adapting to terrain variations.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Aerial additive manufacturing

Early work on aerial additive manufacturing showed that fleets of unmanned aerial vehicles (UAVs) can deposit foam or cementitious materials while flying using dedicated “BuilDrones” for extrusion and “ScanDrones” for in-loop geometric feedback and adaptive path planning. These studies demonstrated multi-robot coordination with model predictive control (MPC) and real-time quality assessment but remained at demonstrator scale with prints limited to small cylinders and dome-like forms. The focus was on proving feasibility rather than delivering structurally validated, field-deployable elements.

More recent work proposes a fully autonomous chunk-based Aerial-AM framework that decomposes complex meshes into printable sub-components linked by a dependency graph. This allows a UAV to construct large geometries in smaller, manageable chunks while an offset-free model-predictive controller compensates for ground effect and disturbances during flight. However, the framework still relies exclusively on UAVs, uses lightweight expanding foam as the primary material and does not provide quantitative structural performance data or address limitations in reach, payload and precision for building-scale components.

Complementary studies on extrusion-based aerial AM focus on detailed flight path tracking and deposition behaviour. Velu et al. investigates UAV-based roof repair, showing good agreement between simulated and experimental paths but also highlighting increased horizontal deviation and thinner extrusions as payload and downwash effects grow. Work on pseudoplastic cementitious mortars demonstrates that suitably tuned mixes can be extruded mid-air by untethered UAVs and maintain buildability, yet long-term structural performance and durability of aerially printed elements remain largely unexplored. These works confirm that aerial robots can deposit both foam and cementitious materials but they leave open questions on strength, service life and robustness in real construction environments.

2.2 Materials and sustainability in construction 3D printing

Parallel literature on terrestrial 3D printing of buildings reports significant economic and environmental advantages over conventional construction. Reviews of large-scale concrete printing conclude that printed structural and envelope components can reduce material waste, labour requirements and construction time while offering comparable or lower lifecycle costs. Demonstrations in India show that 3D-printed single-storey houses can be erected in a few days with cost per square foot competitive with traditional masonry. At the same time, authors note unresolved issues around interoperability between architectural design tools, structural analysis software and printer control systems, which complicates end-to-end digital workflows.

In the aerial domain, recent reviews of material systems for Aerial-AM emphasise the need for rapid-setting, low-density foams and rheology-controlled mortars that can resist both self-weight and rotor downwash while curing. These reviews argue that aerial additive manufacturing can, in principle, meet structural performance requirements if materials are carefully formulated, but they also acknowledge that tested structures are small and non-habitable, with limited data on behaviour under realistic environmental loads.

2.3 Mobile ground systems, power and logistics

On the ground side, mobile robotic platforms have been proposed to overcome the footprint limitations of gantry-based printers. Sustarevas et al. present an autonomous mobile manipulator that performs large-scale 3D printing by coordinating base motion with arm trajectories using task-consistent path planning and model-predictive control. Their results show accurate nozzle localisation over hundreds of metres of print path but structural load testing is omitted and experiments are limited to controlled indoor environments.

Other researchers use small differential-drive robots coordinated through ROS2 and visual SLAM to perform collaborative sand-deposition tasks, illustrating how multi-robot scheduling and task allocation can extend effective print volume. However, these systems operate with lightweight granular materials on flat lab floors and do not address heavy construction mixes or field conditions.

Power and logistics constraints have motivated work on mobile, solar-powered 3D printers. King and Pearce design open-source, PV-powered FDM printers, one suitcase-scale and one community-scale, demonstrating that photovoltaics can sustain multi-hour printing in off-grid settings. Yet these studies focus on plastic filament for appropriate technology rather than construction materials and they do not consider coordination with UAVs or AGVs.

Collectively, the ground-robotics and solar-printing literature suggests that mobile platforms and renewable power can extend printing reach and autonomy but integration with aerial systems for cooperative construction or for tasks such as in-field recharging and canister replacement is still largely unexplored.

2.4 Identified gaps and research focus

Across aerial and mobile 3D printing three major gaps emerge. First, structural validation is limited: most aerial works report geometric accuracy and visual quality but very few provide compressive strength, stiffness or long-term durability data for printed elements. Second, system-level integration is incomplete: aerial frameworks largely assume UAV-only construction, while ground systems and solar printers treat mobility or power constraints separately without heterogeneous robot collaboration. Third, actuation and control interfaces for construction-grade payloads receive little attention; most studies rely on custom extruders or servo-driven valves with minimal discussion of how standard flight controllers interface with high-current actuators such as solenoids.

The project addresses the third gap directly and contributes to the second: it develops a robust, flight-controller-compatible solenoid actuation circuit for aerial material valves and proposes an AGV that shadows the drone as a mobile recharging and canister exchange station. By ensuring reliable on-board dispensing and simplifying resupply, the system forms a practical backbone for future hybrid aerial-ground construction platforms.

2.5 Societal and industry significance

For industry, such a system targets high-value use cases where rapid, minimally manned construction is essential: emergency shelters after floods or earthquakes, repair of hard-to-reach roofs and facades, and incremental retrofits in constrained urban sites. In the Indian context, 3D-printed housing has already shown potential to reduce construction time and cost by up to 30% compared to conventional methods, while improving safety by removing workers from hazardous tasks. Integrating aerial robots with mobile ground support could further reduce scaffolding, crane use and manual handling of materials, directly aligning with Industry 4.0 goals for automation, safety and sustainability.

Societally, autonomous or semi-autonomous Aerial-AM systems promise faster deployment of shelters and critical infrastructure in regions with limited skilled labour or damaged access roads. Coupling drones with an AGV that handles recharging and pressurised canister replacement reduces downtime and dependence on fixed infrastructure, making such systems attractive for disaster-response agencies and NGOs.

2.6 Alternative technologies and preliminary cost–feasibility

Alternative approaches include large fixed gantry printers, pure UAV-only Aerial-AM or purely ground-based mobile printers. Gantry systems deliver excellent structural quality but are capital-intensive, require prepared sites and are poorly suited to rapidly changing disaster environments. UAV-only systems offer unmatched access but are constrained by payload, flight time and the difficulty of carrying heavy power and material supplies. Ground-only mobile printers mitigate footprint issues but cannot reach roofs or vertical surfaces without additional mechanical aids.

The proposed hybrid drone-AGV concept aims to balance these trade-offs. A single construction-grade UAV with a Pixhawk-driven solenoid valve and pressurised canister, combined with a modest AGV for battery charging and canister handling is expected to cost substantially less than a full gantry system while offering higher deployment flexibility. Previous cost studies of printed houses in India suggest that once the equipment is procured, the per-unit construction cost can be brought below that of traditional methods, primarily through reduced labour and schedule

compression. By reusing commercially available components and open-source software (ArduPilot), the actuation architecture developed in this project is both technically feasible and economically attractive as a building block for scalable aerial-ground construction solutions.

Parameter	**Fixed Gantry 3D Printer**	**Pure UAV-Based Aerial AM**	**Proposed AGV + Drone Hybrid**
Mobility	Static (fixed footprint)	High (complete aerial coverage)	High (AGV navigates terrain, drone provides feedback)
Deployment Speed	Slow (3–7 days site setup)	Fast (1–2 days)	Moderate (2–4 days initial setup)
Structural Capacity	Very High (concrete, heavy loads)	Low-Medium (foam, light composites)	Medium-High (PU foam to mortar, scalable)
Cost (Prototype Scale)	₹12–20 lakhs	₹5–8 lakhs	₹6–9 lakhs
Power Independence	Low (requires grid/generator)	Medium (battery-limited 30–45 min)	High (solar assist extends 2–4 hours)
Suitability for Remote/Emergency	Poor	Medium (flight time limited)	Excellent (mobile, durable, testable)

Table 2.1: Comparison chart of alternate technologies

CHAPTER 3

PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Definition

Current aerial additive manufacturing research demonstrates impressive geometric capabilities but reveals a critical gap in practical field deployment, i.e, reliable material dispensing control during untethered flight. While chunk-based frameworks, extrusion path planning and pseudoplastic mortar formulations show theoretical feasibility, no standardized interface exists between commercial flight controllers like Pixhawk 2.4.8 and construction-grade solenoid actuators needed for precise valve operation.

The current state features sophisticated aerial path planning and multi-robot coordination, yet material release remains either manually triggered, custom-controller dependent or limited to lab-scale demonstrations. The desired future state requires drones to autonomously deposit PU foam or mortar in layers during real construction missions, synchronized with AGVs providing mobile battery charging and canister exchange. The gap lies in translating Pixhawk's 3.3V PWM servo signals into stable digital ON/OFF control for suitable push-pull solenoids, ensuring repeatable actuation without partial strokes, voltage leakage, or flight safety risks that plague direct connections.

3.2 Objectives

The main objectives of this project are:

- To design and develop an Autonomous Ground Vehicle (AGV) capable of wireless control and basic material carrying functionality.
- To assemble and configure a quadcopter drone using Pixhawk flight controller and QGroundControl for manual operation and calibration.
- To design and implement a lightweight material dispensing mechanism using a canister and solenoid valve system.
- To develop a control interface for the AGV using ESP32 and a web-based platform for real-time operation.

- To test the functionality of individual subsystems, including AGV movement, drone configuration and controlled material dispensing.
- To analyze the limitations of the current prototype and identify improvements required for achieving a fully integrated collaborative system. The work is limited to prototype-level implementation, including AGV control, drone configuration and material dispensing mechanism.
- Full system integration, autonomous coordination and aerial material deposition are considered as future work due to practical constraints such as stability and payload limitations.

3.3 Sustainability Goals

Our project connects directly to UN SDG 9 (Industry, Innovation, and Infrastructure) by creating robotic tools that can quickly build essential shelters in disaster areas. The drone-AGV team works much faster than traditional methods and needs far fewer workers, helping communities recover when roads are blocked and help is slow to arrive.

It supports SDG 11 (Sustainable Cities and Communities) by using materials more efficiently and eliminating the need for cranes, scaffolding and heavy equipment that damage construction sites and create waste.

For SDG 13 (Climate Action), the solar-powered AGV keeps the drone flying longer without relying on diesel generators or grid power, cutting carbon emissions during emergency response operations.

Finally, SDG 7 (Affordable and Clean Energy) benefits from our mobile charging system that makes the most of sunlight even in remote areas, ensuring continuous operation where power infrastructure doesn't exist.

CHAPTER 4

METHODOLOGY

4.1 Overview

The proposed system consists of two primary subsystems: an Autonomous Ground Vehicle (AGV) and a quadcopter drone. The AGV is designed to function as a mobile platform for movement and potential material support, while the drone is developed for controlled material dispensing. The methodology focuses on the design, development and testing of individual subsystems rather than full system integration. The objective is to evaluate the feasibility of combining ground and aerial platforms for assisted construction applications.

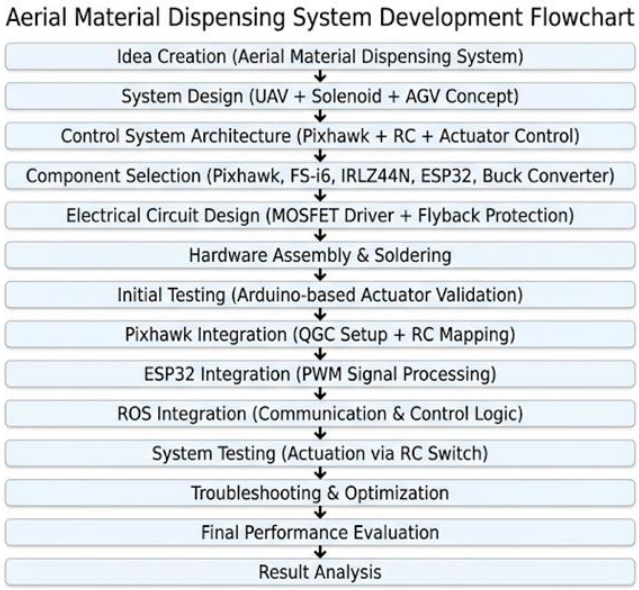


Figure 4.1: Methodology flowchart

4.2 Materials and Equipment

The project utilized commercially available components optimized for UAV integration with the core components being:

- JF-1039B Push-Pull Solenoid: 6V DC, 25N force, 10mm stroke for precise canister valve actuation
- Pixhawk 2.4.8 Flight Controller: 32-bit ARM Cortex-M4 processor, 8-16 PWM outputs, 3.3V logic levels
- IRLZ44N N-Channel MOSFET: Logic-level switching, Vds=55V, Id=47A, Rds(on)=22mΩ at Vgs=5V

- MP1584EN DC-DC Buck Converter: 4.5-28V input, 0.8-20V adjustable output, 3A continuous current
- 1N5408 Flyback Diode: 1000V PIV, 3A continuous, 200A surge rating
- AGV Platform: Four-wheel drive mobile base, DC motors with L298N motor driver, powered by a 24V / 12Ah battery, integrated with ESP32 control node for navigation and coordination
- S500 Quadcopter Frame: 480mm motor-to-motor, 1.5-2kg payload capacity
- FlySky FS-i6 Transmitter/Receiver: 6-channel 2.4GHz, PPM/SBUS output
- Power System: 3S LiPo battery (11.1V nominal, 2200mAh), stepped down to 6V for solenoid operation.

Each component was selected carefully by referring to the data sheets and ensuring the compatibility with one another.

4.3 Development of the system

The development followed an iterative test cycle, progressing from concept to flight-ready integration:

Step 1: System Conceptualization

The need for Pixhawk-compatible solenoid control enabling transmitter-triggered material dispensing was identified. Defined requirements: clean 6V ON/OFF switching from 3.3V PWM AUX signals, less than 50g added weight, reliable operation under vibration.

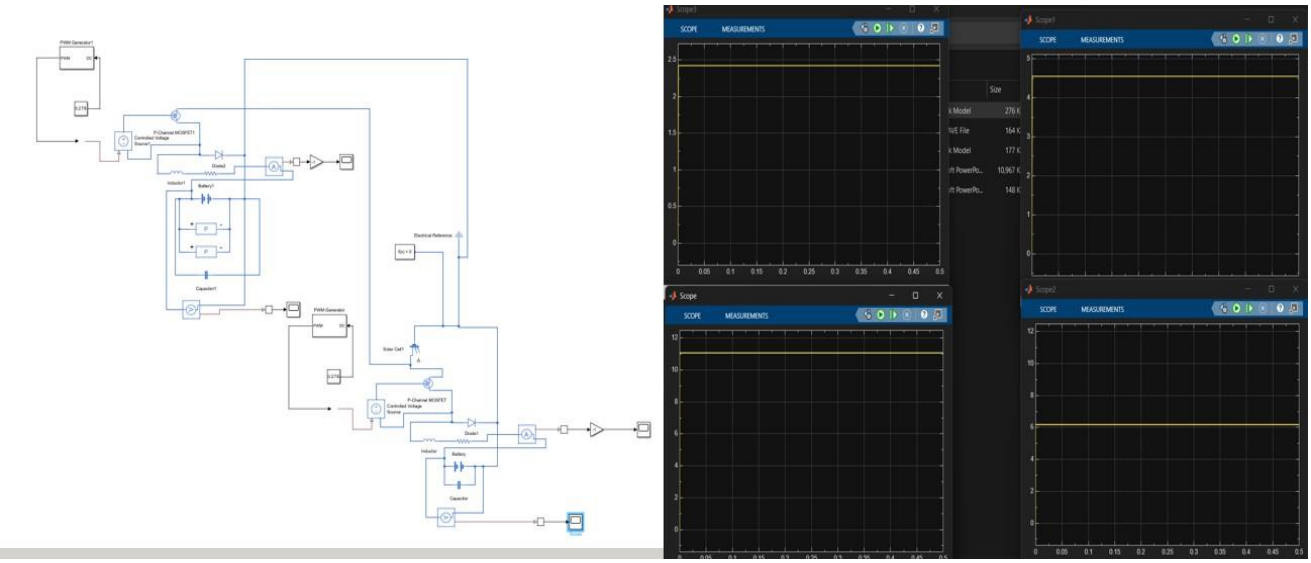


Figure 4.2: Simulink model of power distribution from solar to power module

Step 2: Control Architecture Design
We selected MOSFET low-side switching topology to interface Pixhawk AUX1 output with solenoid. The gate drive equation ensured reliable switching:

$$V_{\{GS\}} = V_{\{PWM\}} \times \frac{R_{\{pull-down\}}}{R_{\{series\}} + R_{\{pull-down\}}}$$

To estimate material flow rate and structural layer thickness, the following governing relationships were used:

$$Q = A \times v$$

Where Q=Volumetric flowrate (m^3/s),
A=Nozzle cross-sectional area (m^2),
v= Exit velocity of the material (m/s).

For each layer height hand deposition speed s, the mass flow was calculated as:

$$\dot{m} = \rho\ A\ v = \rho\ s\ h\ w$$

Where ρ= density of material, w= layer width.

These relations helped optimize the drone's printing velocity and material discharge rate to maintain consistent layer thickness. These equations were used as a reference for understanding material flow behaviour. However, detailed optimization was not performed in this prototype stage.

Step 3: CAD Modeling
We designed a payload nozzle extruder in SolidWorks for S500 frame compatibility. Optimized center-of-gravity placement and ensured solenoid alignment with canister valve. 3D-printed prototypes validated fitment before final fabrication. We also modelled a material holding structure for the AGV, for the drone to land and exchange cannisters.

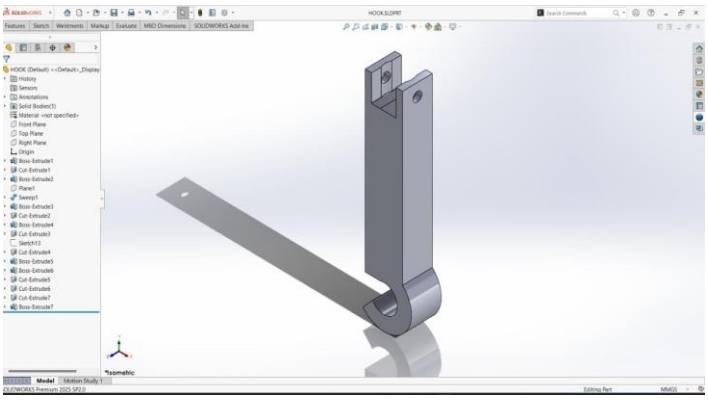


Figure 4.3: CAD model of the nozzle connector

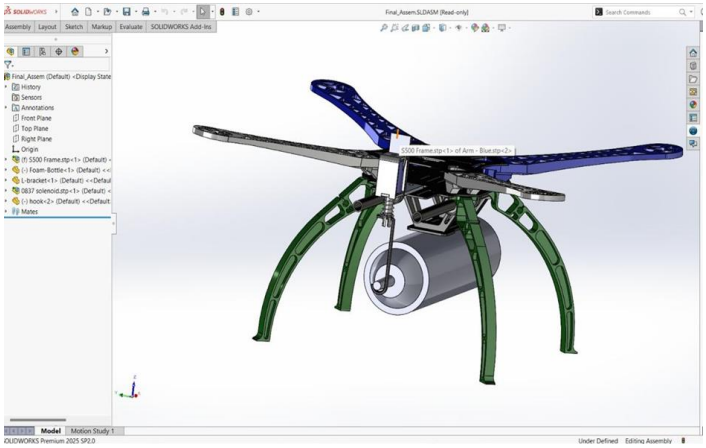


Figure 4.4: CAD assembly of the system

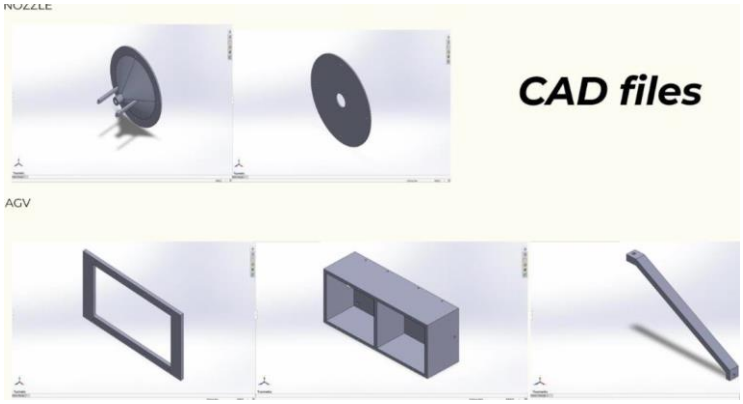


Figure 4.5: CAD model of AGV structure and nozzle

Step 4: Electrical System Design and Assembly
The soldered circuit confirmed basic switching:

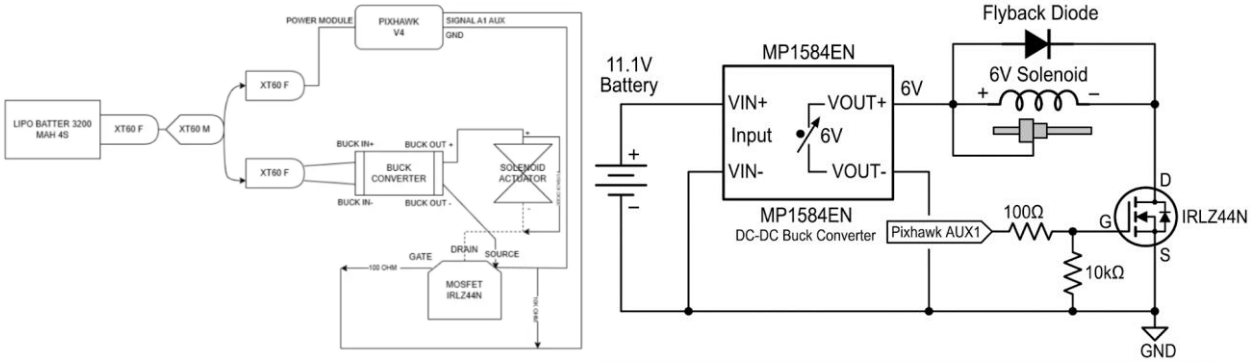


Figure 4.6: Electrical circuit and connections

Components soldered with 22AWG power wiring and strain relief. MP1584EN trimmed to exactly 6.0V under load. A fine-tipped soldering iron, flux, and lead-free solder wire were used to ensure reliable connections. After assembly, the circuit was connected to the power supply and tested for correct voltage levels and functionality of all major components.

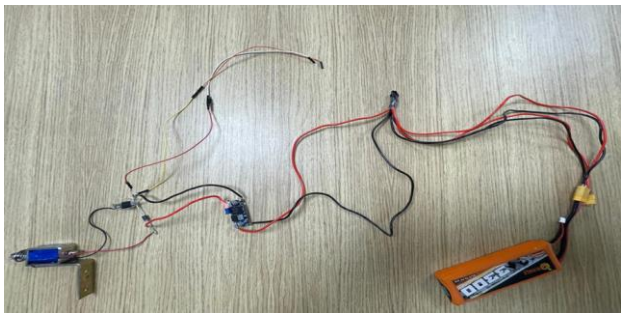


Figure 4.7: Soldered hardware components

Step 5: Software Configuration and Troubleshooting
QGroundControl configured AUX1 as "Relay" function. Radio calibration mapped transmitter switch (Channel 5) to 1000-2000µs PWM range. Initial tests revealed partial actuation due to gate charge timing; resolved by adjusting PWM frequency to 400Hz and adding 100nF gate capacitor to the circuit.

Step 6: Flight-Readiness Validation
Payload mounted on S500 frame. Static hover tests confirmed no control anomalies. Transmitter switch successfully triggered solenoid through full flight envelope. The AGV movement was tested using the web interface, and reliable control was achieved. The drone system was configured successfully; however, stable flight with payload was not achieved. The material dispensing mechanism was tested at ground level by manually moving the drone, demonstrating controlled spraying functionality.

INPUT COMMAND	MOTOR 1	MOTOR 2	MOTOR 3	MOTOR 4	MOTION
Forward	HIGH	HIGH	HIGH	HIGH	Forward
Backward	LOW	LOW	LOW	LOW	Reverse
Left	LOW	HIGH	LOW	HIGH	Turn Left
Right	HIGH	LOW	HIGH	LOW	Turn Right
Stop	LOW	LOW	LOW	LOW	Stop

Table 4.1: Control table

CHAPTER 5

RESULTS AND DISCUSSION

The developed system integrates a UAV-based material dispensing mechanism with a ground-based AGV support unit, demonstrating a coordinated approach toward aerial additive manufacturing. The primary objective was to achieve controlled actuation of a solenoid valve using a flight controller interface, while ensuring system stability, reliability, and scalability.

5.1 Description of the Work Done

The project began with the design and integration of a quadcopter UAV (S500 frame) equipped with a Pixhawk 2.4.8 flight controller. A pressurized canister based dispensing mechanism was mounted on the UAV with a push-pull solenoid acting as the primary control element for material release. The solenoid was interfaced using a MOSFET-based driver circuit powered through a regulated DC-DC buck converter. Initial testing revealed incompatibility between the PWM output of the Pixhawk and the switching requirements of the MOSFET. This resulted in partial actuation, voltage leakage and inconsistent solenoid behaviour. Through iterative debugging and hardware refinement, signal conditioning techniques were implemented to ensure reliable ON/OFF switching. The final system successfully achieved controlled actuation via a transmitter switch (FlySky FS-i6) demonstrating stable extension and retraction of the solenoid. The AGV platform was developed using a four-wheel drive system powered by DC motors and controlled via an ESP32 microcontroller. The AGV was designed to act as a mobile ground station, capable of carrying additional payload and supporting extended UAV operations. Integration with Arduino enabled communication architecture development, allowing for future expansion into autonomous coordination between UAV and AGV.



Figure 5.1: Completed drone attachment

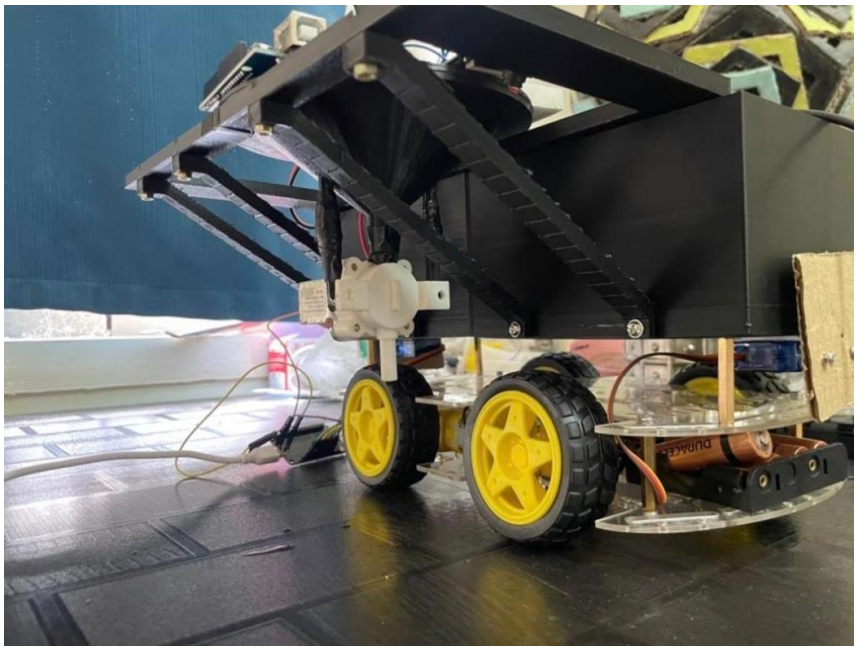


Figure 5.2: AGV ground system

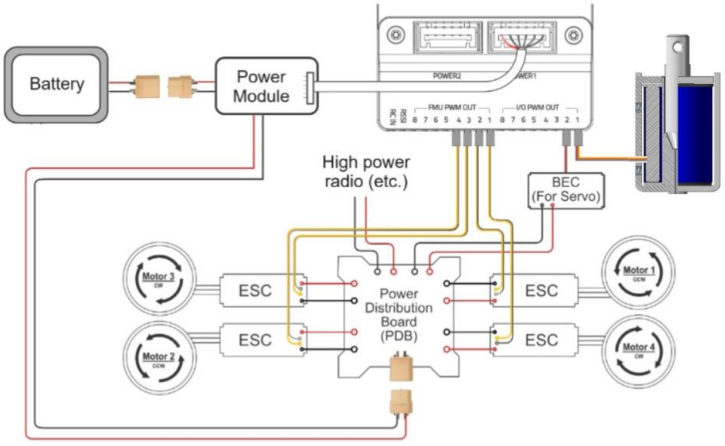


Figure 5.3: Pixhawk to drone connections

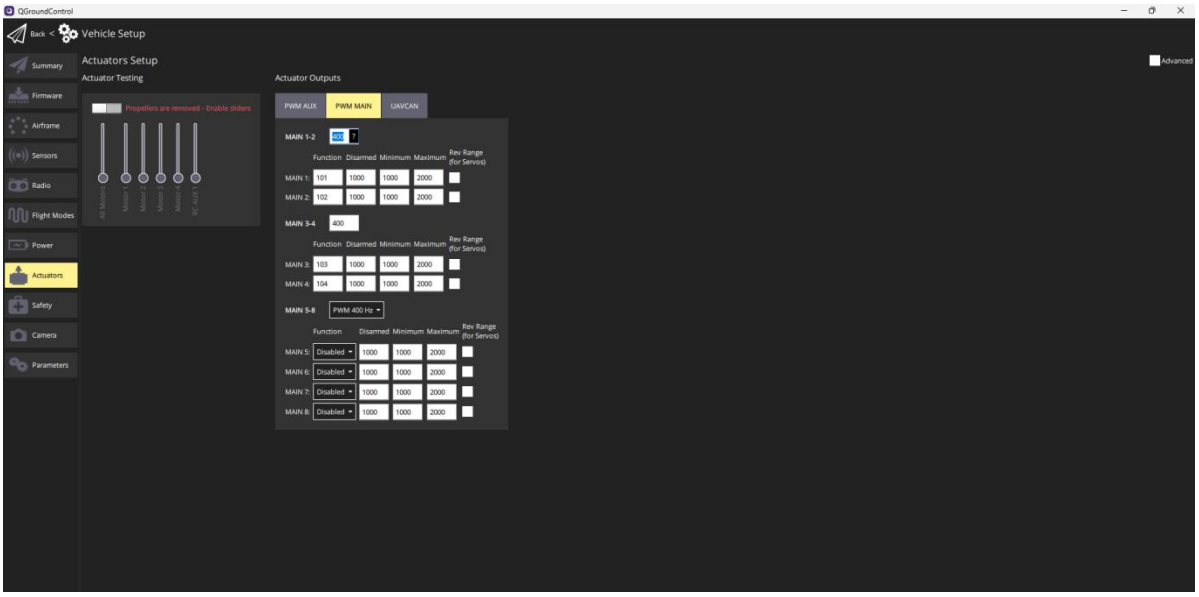


Figure 5.4: QGroundControl motors and actuators calibration

5.2 Results and Performance Analysis

The final system demonstrated the following key results:

- The actuator responded consistently to RC input, achieving controlled material release.
- The buck converter maintained a steady voltage supply, ensuring proper operation of the solenoid without fluctuations.
- The payload system significantly affects flight stability under tested conditions.
- The ground vehicle successfully operated as a mobile support platform with ESP32-based control.
- Experimental observations showed that proper signal conditioning was critical for eliminating partial switching effects. The actuator exhibited full stroke operation only after resolving PWM-induced leakage, validating the importance of interface design between control electronics and power components.

5.3 Discussion and Literature Correlation

The results align with existing research in UAV-based payload delivery and aerial additive manufacturing, where actuator control and signal reliability are identified as critical challenges. Studies indicate that PWM-to-digital signal mismatch often leads to inefficient actuation in embedded systems, which was observed in this project as well.

The use of MOSFET-based switching circuits with proper grounding and flyback protection is consistent with standard practices in power electronics. Additionally, integrating UAV systems with ROS-enabled ground platform reflects current trends in cooperative robotics and multi-agent systems.

5.4 Innovation and Technical Depth

The key innovation of this project lies in the integration of aerial material dispensing with a coordinated ground support system. Unlike conventional UAV payload systems, this approach enables continuous operation through AGV-assisted logistics, making it suitable for scalable construction tasks.

Technical depth was demonstrated through:

- Signal optimization: Resolving PWM compatibility issues for reliable actuation
- Hardware efficiency: Lightweight mounting and compact circuit design
- System integration: Combining UAV, AGV and ESP32
- Modular architecture: Allowing easy expansion for automation and autonomy

5.5 Cost-Benefit and Sustainability Analysis

The prototype was developed using cost-effective, commercially available components. Compared to industrial aerial construction systems, which are significantly expensive, this solution offers a low-cost alternative for research and small-scale deployment.

- Low component cost: Use of off-the-shelf electronics (Pixhawk, ESP32, MOSFETs)
- Reduced operational cost: Minimal human intervention required
- Scalability: System can be expanded with additional UAVs and AGVs

From a sustainability perspective, the system enables material-efficient construction, reducing waste through controlled dispensing.

5.6 Environmental and Societal Impact

The proposed system has strong relevance in disaster response and temporary shelter construction, where rapid deployment is critical. By enabling aerial material deposition, the system can operate in hazardous or inaccessible environments without risking human lives.

Potential impacts include:

- Disaster relief: Rapid construction of emergency shelters
- Remote infrastructure: Deployment in rural or difficult terrains
- Automation in construction: Reducing labor dependency

5.7 Future Scope

Future improvements can include:

- Full autonomous coordination between UAV and AGV using ROS
- Computer vision integration using ESP32-CAM for precision placement
- Enhanced material delivery mechanisms for complex structures
- AI-based path planning and swarm UAV deployment

CHAPTER 6

CONCLUSION

6.1 Core Achievements:

- Successfully developed a Pixhawk-compatible solenoid actuation interface enabling precise material dispensing from transmitter switch commands, meeting the primary objective of reliable UAV valve control.
- Implemented stable 6V switching circuitry with MOSFET low-side drive, flyback protection and optimized signal conditioning, eliminating partial actuation and voltage leakage issues.
- Validated repeatable extension/retraction cycles suitable for layer-by-layer aerial deposition, establishing a re-usable control architecture for both drone and AGV operations.
- Demonstrated bench-tested integration ready for flight, with clean switching behavior across the full PWM range.

6.2 Societal and Industrial Scalability:

The interface design uses commodity components (IRLZ44N MOSFET, MP1584EN buck converters) available globally, enabling mass production at scale. A single PCB could support multiple solenoid channels for complex payload systems. Beyond emergency shelters, the architecture adapts to industrial spraying (pesticides, fire retardants), infrastructure inspection (adhesive markers), and precision agriculture. The AGV-drone model scales to swarm operations where multiple UAVs service from centralized ground stations, dramatically increasing construction throughput while maintaining low per-unit costs.

6.3 Major Technical Hurdles:

Initial Pixhawk PWM signals caused incomplete IRLZ44N switching due to slow gate charge/discharge. Solution: Added 100Ω series resistance and 10kΩ pulldown, plus 100nF gate capacitor, reducing rise/fall times from >200ms to <50ms.

Higher PWM frequencies (>1kHz) caused erratic solenoid behavior. Solution: Fixed Mission Planner output to 400Hz and calibrated transmitter switch mapping (1000-2000μs), ensuring stable digital ON/OFF interpretation.

6.4 Future Technical Upgrades:

The outcomes of this project demonstrate that even simplified prototypes can provide valuable insights into complex engineering systems. By addressing practical challenges and validating core concepts, this work contributes to the ongoing development of autonomous and semi-autonomous construction technologies for real-world applications. Future work can focus on improving and expanding the system in the following areas:

- Multi-channel PCB with integrated current sensing for real-time material flow monitoring and predictive maintenance.
- Wireless feedback telemetry from solenoid position sensors to MAVLink, enabling closed-loop flow control during autonomous missions.
- Fail-safe latching circuit ensuring valve closure on power loss or signal dropout.

- AGV integration with ROS2 navigation following drone telemetry for autonomous canister exchange.

This work establishes a practical foundation for aerial construction payloads, bridging the gap between research demonstrators and field-deployable systems.

REFERENCES

Journal Publications:

1. Hager, I., Golonka, A. and Putanowicz, R. (2016), '3D printing of buildings and building components as the future of sustainable construction', Procedia Engineering 151, 292–299.
2. Zhang, K., Chermprayong, P., Xiao, F., Tzoumanikas, D., Dams, B., Kay, S., Kocer, B. B., Burns, A., Orr, L., Alhinai, T., Choi, C., Darekar, D. D., Li, W., Hirschmann, S., Soana, V., Ngah, S. A., Grillot, C., Sareh, S., Choubey, A. and Margheri, L. (2022), 'Aerial additive manufacturing with multiple autonomous robots', Nature 609(7928), 709–717.
3. Dams, B., Chen, B., Kaya, Y. F., Orr, L., Kocer, B. B., Shepherd, P., Kovac, M. and Ball, R. (2024), 'Fresh properties and autonomous deposition of pseudoplastic cementitious mortars for aerial additive manufacturing', IEEE Access 12, 34606–34631.
4. King, D., Babasola, A., Rozario, J. and Pearce, J. (2014), 'Mobile open-source solar-powered 3-D printers for distributed manufacturing in off-grid communities', Challenges in Sustainability 2(1), 72–93.
5. Stamatopoulos, M. N. and Haluška, J. (2025), 'Fully autonomous chunk-based aerial additive manufacturing with offset-free predictive control', Automation in Construction 158, 106361.
6. Dams, B., Chen, B., Kaya, Y. F., Shepherd, P., Kovac, M. and Ball, R. (2025), 'The rise of aerial additive manufacturing in construction: A review of material advancements', Construction and Building Materials (in press).

Conference Proceedings:

7. Velu, R., Roy, A., Adhikari, C. S., Kumar, N., Manipuzha, A. S. and Teacher, M. (2025), 'Experimental investigation and simulation on extrusion-based aerial additive manufacturing', Proceedings of the International Conference on Emerging Trends in Automation and Advanced Manufacturing, IEEE, pp. 1–8.
8. Ball, R. J., Dams, B., Shepherd, P. and Chen, B. (2025), 'Exploring aerial additive manufacturing with cementitious materials using autonomous drones', Proceedings of the International Conference on Durability of Concrete Structures, Purdue University Press, pp. 1–10.

9. Sustarevas, J., Kanoulas, D. and Julier, S. (2022), ‘Autonomous mobile 3D printing of large-scale trajectories’, Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, pp. 1–8.
10. Zhou, T. Z., Wang, S. and Sheng, Y. (2025), ‘Collaborative fabrication with multiple robots using ROS2, visual SLAM and Rhino Grasshopper’, Proceedings of the 30th CAADRIA Conference, CAADRIA, pp. 525–534.

Websites:

11. PX4 Development Team. (2025, July 9). ‘Pixhawk 4 wiring quick start’. PX4 User Guide. Retrieved from https://docs.px4.io/v1.13/en/assembly/quick_start_pixhawk4
12. ArduPilot Dev Team. (2024). ‘Pixhawk overview and PWM servo outputs’. ArduPilot Documentation. Retrieved from https://ardupilot.org/copter/docs/common-pixhawk-overview.html
13. Xcluma. (2026). ‘JF 1039B DC 6V 25N 10mm stroke push pull type DC solenoid electromagnet linear motion’. Product datasheet. Retrieved from https://www.xcluma.com/ (search JF 1039B).
14. Robu.in. (2025). ‘Mini MP1584 DC DC 3A adjustable buck module’. Product page. Retrieved from https://robu.in/product/mini-mp1584-dc-dc-adjustable-buck-module-3a/
15. Vishay. (2021). ‘IRLZ44N power MOSFET datasheet’. Vishay Semiconductors. Retrieved from https://www.vishay.com/docs/91328/irlz44.pdf
16. Blikai Electronics. (2025). ‘IRLZ44N MOSFET transistor: specs, features, and applications’. Blikai. Retrieved from https://www.blikai.com/blog/featured-products/irlz44n-mosfet-transistor-specs-features-and-applications
17. The Engineering Projects. (2020). ‘1N5408 power diode: pinout, datasheet, features & applications’. The Engineering Projects. Retrieved from https://www.theengineeringprojects.com/2020/09/1n5408-power-diode-pinout-datasheet-features-equivalents.html
18. ProtoSupplies. (2025). ‘MP1584 mini adjustable DC DC step-down module’. ProtoSupplies. Retrieved from https://protosupplies.com/product/mp1584-mini-adjustable-dc-dc-step-down-module/